

Lakshya JEE 2.0 2025

Maths

DPP: 4

Inverse Trigonometric Functions

Q1 Prove that

$$\sin^{-1} \frac{1}{\sqrt{17}} + \cos^{-1} \frac{9}{\sqrt{85}} = \tan^{-1} \frac{1}{2}$$

Q2 Find x , if $\cos(\sin^{-1} x) = \frac{1}{2}$

Q3 Find x , if $\tan^{-1} x = \sin^{-1} \frac{1}{\sqrt{2}}$

Q4 The value of $\sum_{r=0}^{\infty} \tan^{-1} \left(\frac{1}{1+r+r^2} \right)$ is equal to

- (A) $\frac{\pi}{2}$
 (B) $\frac{3\pi}{4}$
 (C) $\frac{\pi}{4}$
 (D) None of these

Q5 If $a_1, a_2, a_3, \dots, a_n$ is an A.P. with common difference d , then

$\tan$

$$\left[\tan^{-1} \left(\frac{d}{1+a_1 a_2} \right) + \tan^{-1} \left(\frac{d}{1+a_2 a_3} \right) + \dots \right. \\ \left. \dots + \tan^{-1} \left(\frac{d}{1+a_{n-1} a_n} \right) \right] =$$

- (A) $\frac{(n-1)d}{a_1 + a_n}$
 (B) $\frac{(n-1)d}{1+a_1 a_n}$
 (C) $\frac{nd}{1+a_1 a_n}$
 (D) $\frac{a_n - a_1}{a_n + d_1}$

Q6 $\sum_{r=1}^n \tan^{-1} \left(\frac{2^{r-1}}{1+2^{2r-1}} \right)$ is equal to

- (A) $\tan^{-1}(2^n)$
 (B) $\tan^{-1}(2^n) - \frac{\pi}{4}$
 (C) $\tan^{-1}(2^{n+1})$
 (D) $\tan^{-1}(2^{n+1}) - \frac{\pi}{4}$

Q7

$\cot^{-1}(2 \cdot 1^2) + \cot^{-1}(2 \cdot 2^2) + \cot^{-1}$ is
 $(2 \cdot 3^2) + \dots$

equal to:

- (A) $\frac{\pi}{4}$
 (B) $\frac{\pi}{3}$
 (C) $\frac{\pi}{2}$
 (D) $\frac{\pi}{5}$

Q8 The sum of infinite terms of the series

$$\cot^{-1} \left(1^2 + \frac{3}{4} \right) + \cot^{-1} \left(2^2 + \frac{3}{4} \right) \\ + \cot^{-1} \left(3^2 + \frac{3}{4} \right) + \dots$$

- (A) $\pi/4$
 (B) $\tan^{-1} 2$
 (C) $\tan^{-1} 3$
 (D) None of these

Q9 $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{4n}{n^4 - 2n^2 + 2} \right)$ is:

- (A) $\frac{\pi}{4}$
 (B) $\frac{3\pi}{4}$
 (C) $\frac{\pi}{6}$
 (D) $\frac{5\pi}{6}$

Q10 The sum of series

$$\tan^{-1} \left(\frac{4}{13} \right) + \tan^{-1} \left(\frac{6}{73} \right) + \tan^{-1} \left(\frac{8}{241} \right) + \dots \infty$$

- (A) $\cot^{-1}(2)$
 (B) $\tan^{-1}(2)$
 (C) $\frac{\pi}{4}$
 (D) $\frac{\pi}{6}$

Q11 The value of $\sin(2 \cos^{-1} x + \sin^{-1} x)$ when
 $x = \frac{1}{5}$



- (A) $1/5$ (B) $-1/5$
(C) 5 (D) -5

Q12 Suppose

$$3 \sin^{-1}(\log_2 x) + \cos^{-1}(\log_2 y) = \pi/2 \text{ and } \sin^{-1}(\log_2 x) + 2 \cos^{-1}(\log_2 y) = 11\pi/6,$$

then the value of $x^{-2} + y^{-2}$ equals

- (A) 6 (B) 7
(C) 5 (D) $\frac{7}{2}$

Q13 If $a \sin^{-1} x - b \cos^{-1} x = c$, then $a \sin^{-1} x + b \cos^{-1} x$ is equal to:

- (A) 0
(B) $\frac{\pi ab + c(b-a)}{a+b}$
(C) $\frac{\pi}{2}$
(D) $\frac{\pi ab + c(a-b)}{a+b}$

Q14 If $\tan^{-1} x + 2 \cot^{-1} x = \frac{2\pi}{3}$ then x is equal to

- (A) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
(B) 3
(C) $\sqrt{3}$
(D) $\sqrt{2}$

Q15 Number of solutions to the equation

$$\sin^{-1} x - \cos^{-1} x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \text{ is:}$$

- (A) One (B) Two
(C) Four (D) None of these

Q16 If $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$ then x is equal to

- (A) $\frac{7}{13}$
(B) $\frac{4}{3}$
(C) 13
(D) $\frac{13}{7}$

Q17 The number of solutions of the equation

$$\tan^{-1}(1+x) + \tan^{-1}(1-x) = \frac{\pi}{2} \text{ is}$$

- (A) 2 (B) 3
(C) 1 (D) 0

Q18 If $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$, then the value of x is:

- (A) $\frac{3\pi}{4}$
(B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$
(D) None of these

Q19 Number of solutions of the equation

$$\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) \text{ is:}$$

$$= \tan^{-1}\left(\frac{2}{x^2}\right)$$

- (A) 1 (B) 2
(C) 3 (D) 4

Q20 The sum of the solutions of the equation

$$2 \sin^{-1} \sqrt{x^2 + x + 1} + \cos^{-1} \sqrt{x^2 + x} = \frac{3\pi}{2}$$

- (A) 0 (B) -1
(C) 1 (D) 2

Q21 Solution of the equation

$$\cot^{-1} x + \sin^{-1} \frac{1}{\sqrt{5}} = \frac{\pi}{4} \text{ is:}$$

- (A) $x = 3$
(B) $x = \frac{1}{\sqrt{5}}$
(C) $x = 0$
(D) None of these

Q22 If $\cos^{-1} x > \sin^{-1} x$, then:

- (A) $x < 0$
(B) $-1 < x < 0$
(C) $0 \leq x < \frac{1}{\sqrt{2}}$
(D) $-1 \leq x < \frac{1}{\sqrt{2}}$



Answer Key

Q1 Hence proved.

Q2 $x = \pm \frac{\sqrt{3}}{2}$

Q3 1

Q4 (A)

Q5 (B)

Q6 (B)

Q7 (A)

Q8 (B)

Q9 (B)

Q10 (A)

Q11 (A)

Q12 (A)

Q13 (D)

Q14 (C)

Q15 (A)

Q16 (C)

Q17 (C)

Q18 (B)

Q19 (A)

Q20 (B)

Q21 (A)

Q22 (D)



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